

Supplemental Material:
**A characterization of discrete homotopy distance in complex
networks**

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Abstract

This Supplemental Material provides complete proofs of the two theorems stated in the main text. We fix notation and record the elementary lower bound (Sec. S1), prove that the interleaving criterion and its discrepancy-set reformulation are equivalent (Sec. S2), and give the full constructive proof of the characterization (Sec. S3). The proof is organized around the observation that the interleaving condition orients the “internal edges” of the discrepancy set; because these edges form paths, the induced constraints are acyclic and admit a valid correction order. Finally we show that the argument uses only the covering relations along the two chains and therefore extends verbatim from graded posets to arbitrary saturated chains of equal length (Sec. S4).

S1. PRELIMINARIES AND NOTATION

Let P be a poset and let $x \prec y$ denote the covering relation: $x < y$ and there is no z with $x < z < y$. A chain $x_0 < x_1 < \cdots < x_m$ is *saturated* if $x_k \prec x_{k+1}$ for every k ; a *maximal chain* is a saturated chain from a minimal to a maximal element of P .

Throughout, F and G are saturated chains of the same finite length n that share their endpoints,

$$F = (z_0^F \prec \cdots \prec z_n^F), \quad G = (z_0^G \prec \cdots \prec z_n^G), \quad (\text{S1})$$

with $z_0^F = z_0^G$ and $z_n^F = z_n^G$. (In a graded poset P with a unique minimum and maximum—in particular in $\mathcal{H}(N)$ [2]—every pair of maximal chains is of this form.) We write

$$D = \{i : z_i^F \neq z_i^G\}, \quad m_D = \max D, \quad (\text{S2})$$

so that $D \subseteq \{1, \dots, n-1\}$ and $0, n \notin D$. For a saturated chain H and an element x we write $H(z_\ell^H \rightarrow x)$ for the sequence obtained from H by replacing its rank- ℓ entry with x .

We say indices k and $k+1$ span an *internal edge* of D if $\{k, k+1\} \subseteq D$. For each k define the two *cross-cover* predicates

$$A_k : z_k^F \prec z_{k+1}^G, \quad B_k : z_k^G \prec z_{k+1}^F. \quad (\text{S3})$$

Definition 1 (interleaving). F and G are *interleaved* if $A_k \vee B_k$ holds for every $k \in \{0, \dots, n-1\}$.

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Lemma S1 (single-move validity). *Let H be a saturated chain of length n sharing its endpoints with G , and let $\ell \in \{1, \dots, n-1\}$ with $z_\ell^H \neq z_\ell^G$. If*

$$z_{\ell-1}^H \prec z_\ell^G \prec z_{\ell+1}^H, \quad (\text{S4})$$

then $H' = H(z_\ell^H \rightarrow z_\ell^G)$ is a saturated chain of length n , and H, H' are adjacent (they differ in exactly position ℓ).

Proof. H' differs from H only at position ℓ , so it differs in exactly one position and its endpoints are unchanged. Every consecutive pair of H' other than $(\ell-1, \ell)$ and $(\ell, \ell+1)$ is a cover because it is a cover in H ; the two remaining pairs are covers by (S4). Hence H' is saturated, and a saturated sequence is a chain. Adjacency is Definition 1 of the main text. $\square$

S2. EQUIVALENCE OF THE TWO FORMULATIONS

The main text states the characterization first as interleaving (Theorem 1) and then in the discrepancy-set form used for computation (Theorem 2). We show the two conditions coincide.

Proposition S1. *The following are equivalent:*

- (i) F and G are interleaved (Definition 1);
- (ii) $F = G$, or $|D| = 1$, or $A_k \vee B_k$ holds for every $k \in D \setminus \{m_D\}$.

Proof. We first note that $A_k \vee B_k$ holds automatically unless k and $k+1$ are both in D . Indeed:

- if $k \notin D$, then $z_k^F = z_k^G$; if moreover $k+1 \notin D$ then $z_{k+1}^G = z_{k+1}^F$ and A_k reads $z_k^F \prec z_{k+1}^F$, true because F is saturated; if instead $k+1 \in D$ then B_k reads $z_k^G = z_k^F \prec z_{k+1}^F$, again true;
- if $k \in D$ but $k+1 \notin D$, then $z_{k+1}^G = z_{k+1}^F$ and A_k reads $z_k^F \prec z_{k+1}^F$, true.

Thus the only indices at which $A_k \vee B_k$ can fail are the internal edges of D , and every internal edge k satisfies $k \in D$ and $k+1 \in D$, hence $k \leq m_D - 1$, so $k \in D \setminus \{m_D\}$.

(i) $\Rightarrow$ (ii): interleaving asserts $A_k \vee B_k$ for all k , in particular for all $k \in D \setminus \{m_D\}$.

(ii) $\Rightarrow$ (i): if $F = G$ or $|D| = 1$ there are no internal edges, so by the paragraph above $A_k \vee B_k$ holds for every k . Otherwise (ii) gives $A_k \vee B_k$ for $k \in D \setminus \{m_D\}$, which includes every internal edge; at all other k the disjunction holds automatically. Hence F, G are interleaved. $\square$

Because of Proposition S1 it suffices to prove the characterization in the interleaving form; the discrepancy-set statement then follows.

S3. PROOF OF THE CHARACTERIZATION

Theorem S1 (main text, Theorems 1 and 2). $d_{\mathcal{H}}(F, G)$ equals $|D|$ if and only if F and G are interleaved.

The lower bound $d_{\mathcal{H}}(F, G) \geq |D|$ is established in [1]. It remains to prove that interleaving is sufficient (Sec. S3 B) and necessary (Sec. S3 C) for equality.

A. Orientation of internal edges

Fix an internal edge k (so $k, k+1 \in D$). During any position-by-position correction of F into G —in which each discrepant position is set once, from its F -value to its G -value—consider the two moves that touch positions k and $k+1$. By Lemma S1, setting position $k+1$ to z_{k+1}^G requires its left flank to cover it; if position k has not yet been corrected, that flank is z_k^F , and the requirement is exactly A_k . Symmetrically, setting position k to z_k^G requires its right flank to be covered; if $k+1$ has not yet been corrected, that flank is z_{k+1}^F , and the requirement is exactly B_k . Hence:

$$\begin{aligned} \neg A_k &\Rightarrow k \text{ must be corrected before } k+1, \\ \neg B_k &\Rightarrow k+1 \text{ must be corrected before } k. \end{aligned} \tag{S5}$$

Interleaving ($A_k \vee B_k$) guarantees that at most one of the two implications in (S5) fires, so each internal edge imposes *at most one* precedence constraint between its endpoints.

B. Sufficiency

Assume F, G interleaved. The discrepant positions D decompose into maximal runs of consecutive integers; distinct runs are separated by at least one agreement position, and a

correction at a position with an agreement neighbor imposes no constraint on the neighbor (its flank is a shared G -value, covered by saturation of G). It therefore suffices to order the corrections within a single run $R = \{a, a+1, \dots, b\} \subseteq D$, with $a-1, b+1 \notin D$.

The precedence constraints (S5) within R are carried by its internal edges $a, a+1, \dots, b-1$, each contributing at most one directed constraint between consecutive elements. These constraints live on the path $a - (a+1) - \dots - b$. A path is a tree, and *any* orientation of the edges of a tree is acyclic; hence the constraint digraph on R has no directed cycle. A finite acyclic digraph always has a source—a vertex with no incoming arrow—so at every stage some position of R has all of its forced predecessors already corrected and can be processed next; equivalently, the digraph admits a topological order π of its elements. This is exactly the guarantee that Algorithm 2 of the main text never stalls: a position with valid flanking covers is always available until R is exhausted (Fig. S1).

Correct the positions of R in the order π . We claim every move is valid in the sense of Lemma S1. Consider the move at position $\ell \in R$, when the current chain H holds G -values at the positions already corrected (and at all $i \notin D$) and F -values elsewhere. Its flanks:

- Left flank $z_{\ell-1}^H$. If $\ell = a$ or $\ell - 1 \notin D$ or $\ell - 1$ is already corrected, then $z_{\ell-1}^H$ is a G -value (or the shared endpoint value), and $z_{\ell-1}^H \prec z_\ell^G$ because G is saturated. Otherwise $\ell - 1 \in R$ is uncorrected; since π places ℓ after all its forced predecessors, the edge $(\ell - 1, \ell)$ did not force “ $\ell - 1$ before ℓ ”, so by (S5) $A_{\ell-1}$ holds, i.e. $z_{\ell-1}^F \prec z_\ell^G$, and $z_{\ell-1}^H = z_{\ell-1}^F$.
- Right flank $z_{\ell+1}^H$. If $\ell = b$ or $\ell + 1 \notin D$ or $\ell + 1$ is already corrected, then $z_{\ell+1}^H$ is a G -value and $z_\ell^G \prec z_{\ell+1}^H$ by saturation of G . Otherwise $\ell + 1 \in R$ is uncorrected; since π did not force “ $\ell + 1$ before ℓ ”, by (S5) B_ℓ holds, i.e. $z_\ell^G \prec z_{\ell+1}^F$, and $z_{\ell+1}^H = z_{\ell+1}^F$.

In every case the flanking covers (S4) hold, so by Lemma S1 the move is a valid adjacency and preserves saturation.

Applying this to every run and concatenating, we obtain a discrete homotopy $F = H_0, \dots, H_{|D|} = G$ that corrects each discrepant position exactly once. Its length is $|D|$, so $d_{\mathcal{H}}(F, G) \leq |D|$; therefore, $d_{\mathcal{H}}(F, G) = |D|$. $\square$

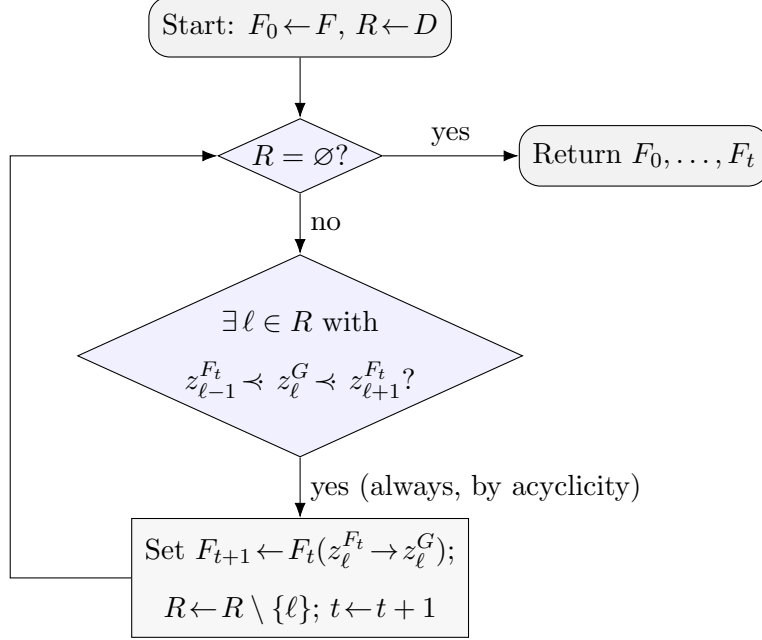


FIG. S1. Decision loop of Algorithm 2 (sufficiency). Under interleaving the constraint digraph on the discrepant positions is acyclic (Sec. S3 B), so a position ℓ with valid flanking covers always exists while $R \neq \emptyset$; the “no” branch of the second test is therefore never taken, and the loop terminates after exactly $|D|$ updates.

C. Necessity

We prove the contrapositive: if F, G are *not* interleaved, then $d_{\mathcal{H}}(F, G) > |D|$. By Proposition S1 the failure occurs at an internal edge k , so $k, k+1 \in D$ and both A_k and B_k fail:

$$z_k^F \not\prec z_{k+1}^G \quad \text{and} \quad z_k^G \not\prec z_{k+1}^F. \quad (\text{S6})$$

Suppose, for contradiction, that $d_{\mathcal{H}}(F, G) = |D|$, and fix a homotopy $F = H_0, \dots, H_{|D|} = G$ realizing it. By the equality clause of Lemma ??, every position of D —in particular k and $k+1$ —is altered exactly once, and no position outside D is ever altered. Thus position k is set once, at some step s , from z_k^F to z_k^G , and position $k+1$ is set once, at some step $t \neq s$, from z_{k+1}^F to z_{k+1}^G .

Consider step s . It replaces $z_k^{H_s} = z_k^F$ by z_k^G ; validity requires $z_k^G \prec z_{k+1}^{H_s}$. The value $z_{k+1}^{H_s}$ is either z_{k+1}^F (if $t > s$) or z_{k+1}^G (if $t < s$). If $t > s$ then $z_k^G \prec z_{k+1}^F$, i.e. B_k , contradicting (S6); hence $t < s$. Now consider step t . It replaces $z_{k+1}^{H_t} = z_{k+1}^F$ by z_{k+1}^G ; validity requires $z_k^{H_t} \prec z_{k+1}^G$. Since $t < s$, position k has not yet been corrected at step t , so $z_k^{H_t} = z_k^F$ and

the requirement is $z_k^F \prec z_{k+1}^G$, i.e. A_k , again contradicting (S6). The two conclusions $t < s$ and $t > s$ are incompatible, so no homotopy of length $|D|$ exists and $d_{\mathcal{H}}(F, G) \geq |D| + 1$. $\square$

Together Secs. S3 B and S3 C prove Theorem S1, and Proposition S1 delivers the discrepancy-set form (Theorem 2 of the main text).

S4. EXTENSION TO SATURATED CHAINS IN ARBITRARY POSETS

Nothing in Secs. S1–S3 uses the rank function of P beyond the two structural facts that (a) F and G are saturated chains of the same length sharing their endpoints, and (b) consecutive entries of each chain form covers. Lemma S1, Lemma ??, Proposition S1, and Theorem S1 were all stated and proved under exactly these hypotheses. We record the resulting generality.

Proposition S2 (Remark 1 of the main text). *Let P be an arbitrary poset and let F, G be saturated chains of the same finite length sharing their endpoints. Then $d_{\mathcal{H}}(F, G)$ equals the number of positions in which F and G differ if and only if F and G are interleaved. In particular, the characterization is not restricted to graded posets; the grading of $\mathcal{H}(N)$ enters only to guarantee, via a unique minimum and maximum, that any two maximal chains automatically satisfy hypotheses (a)–(b) [2].*

Proof. Immediate from Theorem S1, whose proof invokes only (a) and (b). $\square$

This is the sense in which the criterion is genuinely local: it inspects only the covering relations along the two given chains and the finitely many cross-covers (S3), and never the ambient poset. Algorithm 1 of the main text evaluates these on the discrepancy set in $O(n)$ cover queries.

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- [1] C. Goodbrake, D. Beers, T. B. Thompson, H. A. Harrington, and A. Goriely, *Brain chains as topological signatures for Alzheimer’s disease*, J. Appl. Comput. Topol. **8**, 1257 (2024).
 - [2] R. Stanley, *Some aspects of groups acting on finite posets*, J. Combin. Theory Ser. A **32**, 132 (1982).